

FastAPI + SQLAlchemy + Alembic + GCP – Interview Questions

General FastAPI + SQLAlchemy + Alembic

Q: What is FastAPI and why is it used?

A: FastAPI is a modern, high-performance Python web framework for building APIs with automatic Swagger docs, validation, and async support.

Q: How does FastAPI handle dependency injection?

A: FastAPI uses the `Depends` class to manage dependencies injected into routes or services.

Q: What is SQLAlchemy and how does it relate to FastAPI?

A: SQLAlchemy is the ORM used for database interaction in FastAPI. It allows model definitions, queries, and transactions using Python classes.

Q: What is Alembic used for?

A: Alembic handles schema migrations (create, upgrade, downgrade) in SQLAlchemy-backed projects using version-controlled scripts.

Middleware and Swagger

Q: What types of middleware can be added in FastAPI?

A: Logging, CORS, authentication, request/response transformation, rate limiting.

Q: How is Swagger UI configured in FastAPI?

A: FastAPI auto-generates Swagger UI at `/docs`. You can customize it using `FastAPI(title=..., docs_url=..., redoc_url=...)`.

Load Testing (Locust)

Q: What is Locust?

A: Locust is a Python-based load testing tool. It defines user behavior using Python code.

Q: How do you configure Locust for a FastAPI app?

A: Create a `locustfile.py` defining user tasks and target host. Run using `locust -f locustfile.py`.

More in-depth database indexing and performance tuning Q&A;

Q: What are SQLAlchemy indexes and how are they defined?

A: Use `Index()` or `index=True` in model columns. Improves SELECT performance on large tables.

Q: How do you monitor slow queries in PostgreSQL locally?

A: Enable `log_min_duration_statement` in `postgresql.conf` or use `EXPLAIN ANALYZE` for performance debugging.

Q: What is a composite index and how is it created?

A: An index over multiple columns. In SQLAlchemy: `Index('ix_table_col1_col2', Model.col1, Model.col2)`.

GCP-specific deployment or logging questions

Q: How is a FastAPI app deployed on Cloud Run?

A: Package it in a Docker container and deploy using `gcloud run deploy`. Expose port 8080.

Q: How do you connect FastAPI to Cloud SQL (PostgreSQL)?

A: Use a Cloud SQL connector in Python and provide correct connection string, IAM auth or service account credentials.

Q: How do you log structured logs for Cloud Logging (Stackdriver)?

A: Use Python `logging` module with `json.dumps()` format and GCP's logging agent automatically parses it.